[redacted]

MIL-L-45198(Ord)
25 November 1958
SUPERSEDING
PA-PD-64
Dated 11 July 1952

MILITARY SPECIFICATION

LOADING OF HIGH EXPLOSIVE ANTI-TANK SHELL WITH COMPOSITION B

1. SCOPE

1.1 This specification covers the loading of high explosive anti-tank shell with Composition B.

2. APPLICABLE SPECIFICATIONS AND DRAWINGS

2.1 The Ordnance Corps loading drawing pertaining to the shell under contract, and all drawings and specifications referenced thereon, of the issue in effect on the date of invitation for bids, form a part of this specification.

(Copies of drawings and specifications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Materials and parts shall be in accordance with applicable specifications and drawings.

3.1.2 The explosive charge shall comply with all requirements specified on applicable drawings and specifications.

3.1.3 Clean scrap (see 6.1) produced in regular operations may be remelted and reused; however, the scrap shall not be contaminated in any way and shall comply with all requirements for the grade of high explosive specified except for form and viscosity. Scrap from floor sweepings, ventilators, and suction apparatus shall not be used.

FSC 1375

Source: https://assist.dla.mil -- Downloaded: 20261O-22T02:33Z
Check the source to verify that this is the current version before use.

3.2 Specific gravity of the charge. - The specific gravity of the charge shall be not less than 1.65. A lot of shell will be considered to have complied with this requirement if samples selected and tested in accordance with 4.4 comply with the acceptance provisions specified therein.

3.3 Cavities in the charge. - Cavities having a maximum projected dimension of 1/32 inch or less shall be disregarded provided that they are separated by 1/2 inch of solid explosive. Cavities in segment "A" in excess of the size specified in Table I shall be classed as Critical defects. Cavities in segment "B" in excess of the size specified up to and including twice the size specified in Table I shall be classed as Major defects. Cavities in segment "B" in excess of twice the size specified in Table I shall be classed as Critical defects. Cavities in segment "C" in excess of the size specified in Table I shall be classified as Major defects.

3.3.1 Maximum projected area and dimensions of cavities. - Maximum projected area and dimensions of cavities in each of the segments (see figure 1) shall be in accordance with Table I.

TABLE I

	Segment A		Segment B		Segment C
Caliber of Shell	Sum Areas of all cav sq. in	Max. Dimen. of any cav in	Sum Areas of all cav sq in	Max Dimen of any cav in	Sum Areas of all cav sq in
57 mm	1/16	5/16	1/16	1/2	1/16
75 mm	1/16	1/4	1/8	1/2	1/4
90 mm	1/16	3/8	3/16	1/2	1/4
105 mm	1/8	3/8	3/8	5/8	1/4
106 mm	1/8	3/8	3/8	5/8	1/4

3.3.2 Piping cavity. - A piping cavity on the longitudinal axis of the shell between the apex of the cone and the booster well shall be permitted provided that its projected length does not exceed 1 1/2 inches and its width does not exceed 1/8 inch. The projected area of such a cavity shall not be included when determining cavity area in the charge.

3.3.3 Chipping or breaking of charge in booster well and or faced surface of charge.- That portion of Segment A which can be viewed through the opening in the base of the shell shall be inspected visually. Explosive charge chipped or broken from the side wall of the booster well in one or more places shall not be cause for rejection provided that the cavities so formed extend in aggregate around not more than 1/4 of the circumference of the well in the case of the 57mm, 90mm, 105mm and 106mm Shell, and not more than 1/2 of the circumference of the case of the 75mm Shell, and provided that the length of any cavity in the direction parallel to

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:33Z
Check the source to verify that this is the current version before use.

the longitudinal axis of the shell is not more than 1/2 inch and for the 57 and 90mm Shell not more than 1/4 inch, and provided that no cavity extends more than 1/4 inch toward the side wall, and 1/8 inch for 90mm. Cavities in the bottom of the booster well shall not be cause for rejection provided that the sum of the maximum dimensions of the cavities does not exceed 3/16 inch. Cavities having a maximum dimension of 1/32 inch or less shall be disregarded in this computation. Total area of all cavities visible on the faced surface of the charge shall not exceed 1/4 square inch of 1/8 inch depth for the 106mm shell.

3.3.4 Cracks and annular rings. - Cracks shall be permitted if they are no wider than 1/64 inch and if not more than one occurs in any one segment. The presence of an annular ring, 1/16 inch thick maximum and extending from the shell wall 1/4 inch into the charge, shall not be cause for rejection. The cavity area represented by such crack and/or annular ring shall not be included when determining cavity area in the charge. No cracks will be permitted in segment "C".

3.3.5 A lot of shell will be considered to have complied with these requirements if samples selected and examined in accordance with 4.5 comply with the acceptance provisions specified therein. When shell are judged defective on the basis of x-ray examination and are subsequently sectioned, the measurement of cavities visible after sectioning shall not be used as criteria for acceptance or rejection, except as provided in the remainder of the paragraph. In the event x-ray pictures indicate the presence of low density areas and subsequent sectioning shows solid cast throughout the shell except for pinpoint cavities along the shell wall, such cavities shall not be cause for rejection.

3.3.6 The entire cavities extending into two segments shall meet the requirements of both segments.

3.4 Proving ground.

3.4.1 Premature burst. - There shall be no premature burst or evidence thereof (see 6.4).

3.4.2 Functioning. - The shell shall comply with the functioning requirements specified in the respective complete round specification.

3.4.3 A lot of shell will be considered to have complied with these requirements if samples selected and tested in accordance with 4.6 comply with the acceptance provision specified therein.

3.5 Workmanship.

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:33Z
Check the source to verify that this is the current version before use.

3.5.1 The shell shall be free of rust, chips, dirt, grease, and other foreign material. The cleaning method used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used. The interior surface of the shell shall be dry when the charge is loaded into the shell.

3.5.2 Care shall be exercised at all times so that the rotating band of each shell is not damaged by being nicked or burred.

3.5.3 After the booster cavity is formed or drilled, the threads of the shell and the cavity shall be thoroughly cleaned to remove explosive and foreign material. If a chemical cleaner is required to clean the threads, no material other than acetone shall be used. Care should be taken to prevent acetone from coming into contact with the charge and becoming entrapped in the shell.

3.5.4 The base plug or base detonating fuze shall be screwed entirely home and shall bear solidly against the shoulder on the shell and compress the washer next to the charge. The base plug or base detonating fuze shall be caulked in place when required by the loading assembly drawing.

3.5.5 After loading, the exterior of the shell shall be clean and free of explosive.

3.5.6 The painting and markings shall be restored by the loading and assembling contractor on any components furnished to him painted and marked when such paintings and markings have been marred or obliterated.

3.6 Critical defects. -The contractor shall, 100 percent, inspect for each listed critical defect.

4. QUALITY ASSURANCE PROVISIONS

4.1 General Provisions. - The quality assurance provisions contained in this section are applicable to inspection performed by or for the Government and shall be performed as prescribed below.

4.1.1 Contractor quality assurance system. - The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The contractor shall not be restricted to a particular quality assurance system. However, the system adopted must be one which consistently results in material meeting the requirements as evidenced by the evaluation specified in 4.1.2.1. In all testing inspection, the use of the equipment specified and the methods of test prescribed in this specification and other documents referenced herein shall be mandatory. Results of all examinations and tests performed under this quality assurance system shall be made available to the Government. The contractor shall notify the Government of and obtain approval for any change to the written procedure that affects the degree of assurance required by this specification and other documents referenced herein.

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:33Z
Check the source to verify that this is the current version before use.

4.1.1.1 Submission of data. - At the time the product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions and requirements as required by the contractual documents.

b. Number of units of product inspected.

c. Results obtained for all inspections performed.

d. Drawing and specification numbers and dates, together with an identification and date of changes.

e. Number of items in lot.

f. Certificates of analysis including results of tests on materials used in the lot.

g. Date submitted.

4.1.1.2 The certificate shall be signed by a responsible agent of the certifying organization and shall be substantiated by evidence of the agent's authority to bind his principal.

4.1.2 Government verification.

4.1.2.1 At the beginning of the contract and until such time as the contractor has operated under an established and approved quality assurance system which has resulted in an acceptable product, Government verification will consists of (1) performance of all inspections and tests listed in the detail specification and other documents referenced therein, which are not reserved for sole performance by the contractor, (2) monitoring of those inspections reserved for sole performance by the contractor, and (3) monitoring of the contractor's quality assurance operations. At no time will complete Government verification be reduced less than that permitted by MIL-STD-105A and ORD-M608-11 except under those conditions prescribed in 4.1.2.2.

4.1.2.2 Under the following conditions further reduced Government inspection may be initiated on a complete or individual operation basis:

a. On a voluntary basis the contractor includes in his quality assurance provisions written quality control procedure evaluated and approved by the Government.

b. Verification inspection by the Government in accordance with 4.1.2.1 indicates that the quality of product continues to be equal to or better than that required by this specification and the referenced documents.

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:33Z
Check the source to verify that this is the current version before use.

4.1.2.3 At such time as the contractor meets the conditions of 4.1.2.2 above, Government verification will consist of, as a minimum, inspection for critical defects, monitoring of contractor's quality assurance system for strict compliance to the written plan and review of contractor records on all inspection performed. If the contractor's quality deteriorates or a departure from the written plan is noted, the Government inspector will immediately notify the contractor to take corrective action and until the deficiencies have been corrected, Government verification in accordance with 4.1.2.1 will be performed on those characteristics involved to determine acceptability of the product. If corrective action is not taken by the contractor, as requested by the Government, acceptance of the product may be suspended. Upon correction of the deficiencies, the Government will continue inspection as prescribed in 4.1.2.1 until contractor has re-qualified for reduced Government verification as prescribed in paragraph 4.1.2.2.

4.1.2.4 Critical defect.- Government inspection for critical defects will be conducted at all times in accordance with MIL-G-2550.

4.2 Lot formation.

4.2.1 Lot.- A lot shall consist of all high explosive anti-tank shell of the same caliber loaded under an unchanged process by one loading plant, the minimum lot size being at the discretion of the inspector when not specifically prescribed by the detail specification. Each lot shall contain:

a. Loaded fuzes of the same basic lot number.

b. Shell made by one manufacturer at one plant in accordance with the same drawing, same drawing revision, same specification and same specification revision.

c. Other metal parts made by one manufacturer at one plant in accordance with the same drawing, same revision, same specification and same specification revision.

d. Composition B of one manufacturer from not more than one interfix lot number. The composition of the lot with respect to scrap shall be uniform, maintained with ± 5 percent.

4.2.2 List of Inspection Equipment. - Government verification inspection will be accomplished with the use of gages and equipment that has been certified, by the contracting officer, to the design specified in the List of Inspection equipment, number GL_____.

4.2.2.1 Contractor Inspection Equipment. - In inspecting the suppliers inspection equipment, the Government inspector will determine that the supplier has available and utilizes correctly, gaging, measuring and test equipment of required accuracy and precision and that the instruments are of proper type and range to make measurements of the required accuracy. The supplier will have available a set of master gages, standards, and appropriate instruments for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the supplier and made available for review by the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T02:33Z
Check the source to verify that this is the current version before use.

4.2.2.2 Examination Inspection. - Examination inspection will be performed as specified in Specification MIL-G-2550 and as indicated herein. Samples and acceptance numbers for the inspection specified herein will be in accordance with Standard MIL-STD-105 or Publication ORD-M608-11 as appropriate. If it is desired to utilize plans for individual defects, the defect shall be used as a basis for judgement. When the defect is used rather than the defective as a basis for judgement, the following AQL's will be used:

Major Defects	0.25% for each defect
Minor Defects	0.40% for each defect

The contractor may elect to perform all sampling inspection for a given class on an individual defect basis or on a class basis but he shall not use a combination of these.

4.2.3 Tests. -The tests specified in 4.4 shall be conducted by the contractor as specified in MIL-G-2550 in accordance with the methods specified in paragraph 4.4. The contractor shall select the samples for the test as specified below.

4.2.4 Primer. - The contractor shall visually inspect each primer for one or more flash holes missing, prior to insertion into cartridge case.

4.2.5 Critical defects. - Government verification of listed critical characteristics will at all times be performed as below:

Inspection will be performed by a sampling plan with an AQL of 0.015 percent. Any plan from Standard MIL-STD-105 or Publication ORD-M608-11 may be used. The screening inspection normally required of the contractor in conjunction with plans from ORD-M608-11 or in case of rejection by MIL-STD-105 plans will be in addition to the contractor inspection performed prior to presentation.

4.2.6 Classification of defects. - Inspection for characteristics will be performed as follows: (If a sample does not pass the inspection criteria, the material from which the sample was taken will be rejected; however, the contractor may screen out the defect or defects causing the failure and resubmit the material if he so desires.)

4.3 Sampling.

4.3.1 All samples selected by the inspector for testing as specified in 4.6 shall be selected as prescribed in the respective complete round specification.

4.3.2 Samples for cavity examination. -Sampling for cavity examination shall be in accordance with the following Table II.

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:33Z
Check the source to verify that this is the current version before use.

4.3.2 (Cont'd)

TABLE II

Sub-lot size	Sample size	Column I	Column II
1-800	75*	1	*
801-1300	110	2	1
1301-3200	150	3	2
3201-8000	225	4	3
8001-20,000	300	5	4

* On retest use sample size of 110 instead of 75. Acceptance number on retest to be "1".

4.3.3 Inspection. - Inspection shall be as specified in Specification MIL-G-2550 and as indicated herein.

4.4 Determination of specific gravity.

4.4.1 Procedure. - The inspector shall select and identify one shell per each 1000 shell in the lot. The specific gravity of the charge in each sample shell shall be determined as follows: Weigh each sample shell empty (without explosive) (We). Fill each sample shell with water and weigh (Ww). Empty each shell and thoroughly dry. Place the sample shell in the loading line at the rate of approximately one shell per thousand and fill under conditions identical in all respects with those employed in filling regular shell. Weigh each sample shell filled with explosive and having a finished booster cavity (Wt), then reweigh the same shell after filling the booster cavity with water (Wb). Empty the water and thoroughly dry the cavity. Calculate the specific gravity as follows:

$$\text{Specific gravity} = \frac{Wt-We}{(Ww-We)-(Wb-Wt)}$$

where:

We = weight of empty shell
Ww = weight of water filled shell
Wt = weight of loaded shell with finished booster cavity.
Wb = weight of loaded shell with water-filled booster cavity.

4.4.2 Acceptance. -Each group of 1000 shell shall be considered acceptable provided the specific gravity of the charge in the sample shell is not less than 1.65. If the specific gravity of the charge in the sample shell is less than 1.65, the group of 1000 shell represented by the sample shall be rejected.

4.5 Determination of cavities in charge.

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:33Z
Check the source to verify that this is the current version before use.

4.5.1 Procedure. - A sample of shell shall be selected from each sub-lot (see 6.6) in accordance with Table II. Sample shell shall be subjected to X-ray examination by taking a single picture of each perpendicular to its longitudinal axis. In the event an original x-ray picture leaves doubt as to acceptability, the shell shall be re-x-rayed at an angle 90° from that of the original exposure and in the same plane.

4.5.2 Acceptance. - The sub-lot shall be considered acceptable provided no critical defects are encountered and provided the number of sample shell exceeding the requirements of 3.3.1 is not greater than the number shown for the appropriate sub-lot size in Table II, Column I.

4.5.2.1 Rejection. - The sub-lot shall be rejected subject to correction and resubmission for test as specified in 4.5.3 provided no Critical defects are found and if the number of sample shell exceeding the requirements of 3.3.1 is greater than the number shown for the appropriate sub-lot size in Table II, Column I.

4.5.3 Retest. - A retest as provided for in 4.5.2 may be made if requested by the contractor, using sample shell selected in accordance with Table II. The sub-lot shall be considered acceptable on retest provided the number of sample shell exceeding the requirements of 3.3.1 is not greater than the number shown for the appropriate sub-lot size in Table II, Column II. The taking of a second sample shall be predicated upon corrective action having been performed which, in the opinion of the inspector, shall improve the quality of the resubmitted sublot.

4.6 Proving ground test.

4.6.1 Procedure.

4.6.1.1 Functioning. -Details for conducting this test shall be as specified in the respective complete round specification.

4.6.1.2 Safety. - Details for conducting this test shall be as specified in the respective complete round specification except that dummy fuzes shall be used.

4.6.2 Acceptance. -The lot shall be considered acceptable provided that, in the phase being tested (i.e., functioning and/or safety), the sample shell complied with the requirements of the respective complete round specification. The lot shall be considered eligible for retest or subject to rejection in accordance with details specified in the respective complete round specification.

5. PREPARATION FOR DELIVERY

5.1 Packing, labeling and marking. -Packing, labeling and marking shall be as specified on the applicable drawing.

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:33Z
Check the source to verify that this is the current version before use.

5.2 Data cards. - Data card information shall be as specified in Specification MIL-G-2550.

6. NOTES

6.1 Clean scrap is defined as scrap formed in kettles, pails, risers, and loading machines used in the pouring and cooling operations, and scrap formed in the drilling operation.

6.2 It is advisable not to exceed 120° during the melting of Composition B and not to exceed 105° after the composition is molten in order to avoid decomposition.

6.3 In loading the shell it is advisable to have the temperature of the explosive at the time of pouring not less than 84°C.

6.4 A premature burst is defined as a functioning of the projectile in the bore of the gun or during flight before impact.

6.5 The use of the word "sub-lot" in this specification is for identification of qualities of shell for inspection purposes.

Notice. - This specification, together with specifications and drawings pertaining to it and bearing a "Notice" of similar restrictions, is intended for use only in connection with procurement by the United States Government; and shall not be reproduced either wholly or in part except when authorized in connection with Government procurement, or be used for any other purpose except when specifically authorized by the Chief of Ordnance, Department of the Army.

Custodian:
Army-Ordnance Corps

Preparing Activity:
Ordnance Corps

Source: https://assist.dla.mil -- Downloaded: 20261O-22T02:33Z
Check the source to verify that this is the current version before use.

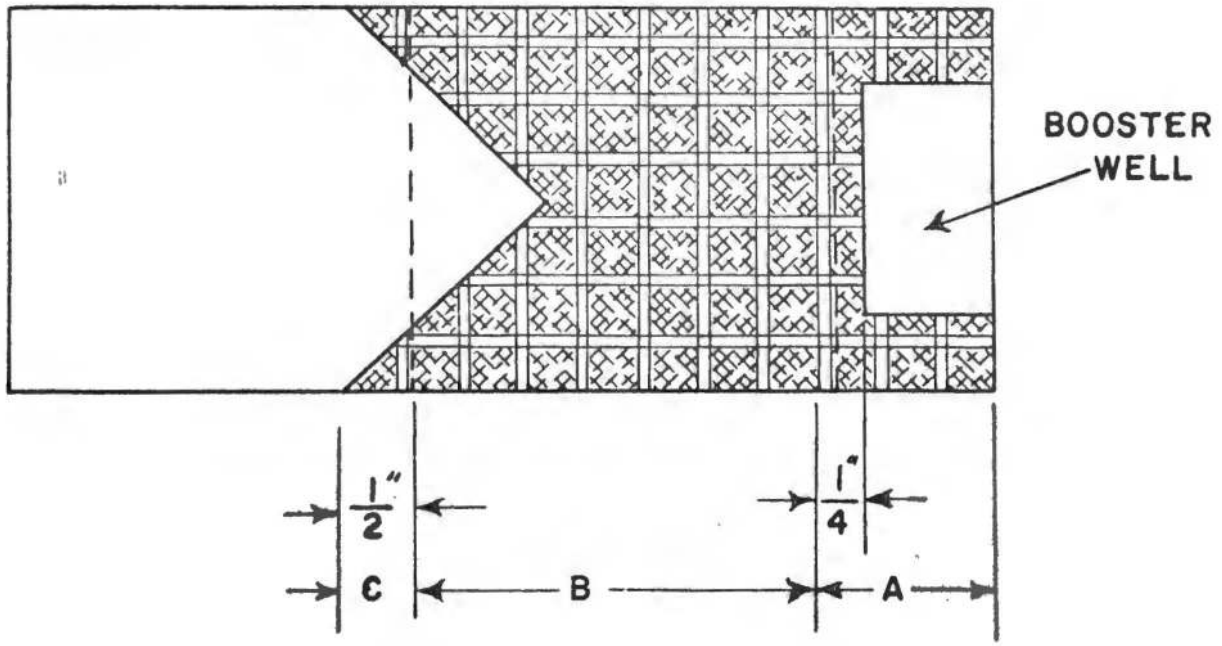


FIGURE 1

DIVISION OF HEAT SHELL INTO SEGMENTS FOR THE PURPOSE OF X-RAY EXAMINATION FOR CAVITIES

Source: https://assist.dla.mil -- Downloaded: 202610-22T02:33Z
Check the source to verify that this is the current version before use.